```
import numpy as np
import pandas as pd
import seaborn as sns
from google.colab import drive

df = pd.read_csv('/content/drive/MyDrive/Dataset.csv')
df.shape
df.info()
df.describe()
df.head()
df.duplicated()
df.duplicated().sum()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   object
 3   Item Purchased          3900 non-null   object
 4   Category                3900 non-null   object
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   object
 7   Size                    3900 non-null   object
 8   Color                   3900 non-null   object
 9   Season                  3900 non-null   object
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   object
 12  Shipping Type           3900 non-null   object
 13  Discount Applied        3900 non-null   object
 14  Promo Code Used         3900 non-null   object
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   object
 17  Frequency of Purchases  3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
np.int64(0)
```

```
df.describe()
```

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

```
cat_col = [col for col in df.columns if df[col].dtype == 'object']
num_col = [col for col in df.columns if df[col].dtype != 'object']
print(cat_col)
print(num_col)
```

```
['Gender', 'Item Purchased', 'Category', 'Location', 'Size', 'Color', 'Season', 'Subscription Status', 'Shipping Type', 'Discoun
['Customer ID', 'Age', 'Purchase Amount (USD)', 'Review Rating', 'Previous Purchases']
```

```
df = df.drop(columns=['Item Purchased','Color','Size'], axis=1)
df.describe(include='all')
```

	Customer ID	Age	Gender	Category	Purchase Amount (USD)	Location	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	P
count	3900.000000	3900.000000	3900	3900	3900.000000	3900	3900	3863.000000	3900	3900	3900	
unique	NaN	NaN	2	4	NaN	50	4	NaN	2	6	2	
top	NaN	NaN	Male	Clothing	NaN	Montana	Spring	NaN	No	Free Shipping	No	
freq	NaN	NaN	2652	1737	NaN	96	999	NaN	2847	675	2223	
mean	1950.500000	44.068462	NaN	NaN	59.764359	NaN	NaN	3.750065	NaN	NaN	NaN	
std	1125.977353	15.207589	NaN	NaN	23.685392	NaN	NaN	0.716983	NaN	NaN	NaN	
min	1.000000	18.000000	NaN	NaN	20.000000	NaN	NaN	2.500000	NaN	NaN	NaN	
25%	975.750000	31.000000	NaN	NaN	39.000000	NaN	NaN	3.100000	NaN	NaN	NaN	
50%	1950.500000	44.000000	NaN	NaN	60.000000	NaN	NaN	3.800000	NaN	NaN	NaN	
75%	2925.250000	57.000000	NaN	NaN	81.000000	NaN	NaN	4.400000	NaN	NaN	NaN	
max	3900.000000	70.000000	NaN	NaN	100.000000	NaN	NaN	5.000000	NaN	NaN	NaN	

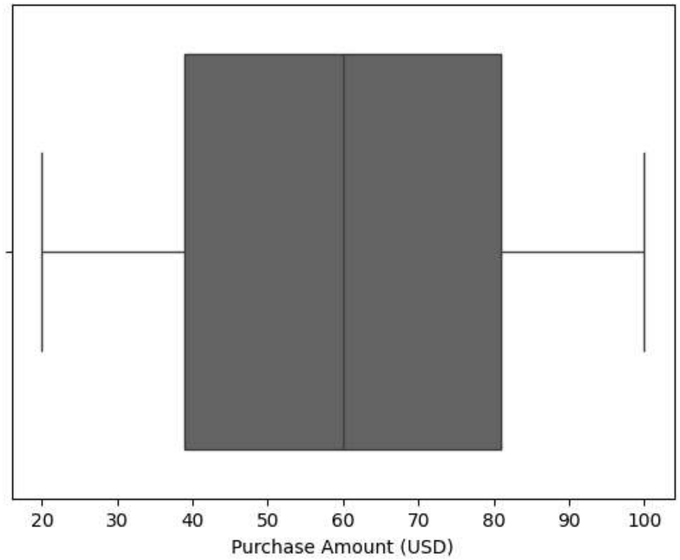
```
df['Review Rating'] = df['Review Rating'].fillna(df['Review Rating'].mean().round(2))
df.isnull().sum()
```

	0
Customer ID	0
Age	0
Gender	0
Category	0
Purchase Amount (USD)	0
Location	0
Season	0
Review Rating	0
Subscription Status	0
Shipping Type	0
Discount Applied	0
Promo Code Used	0
Previous Purchases	0
Payment Method	0
Frequency of Purchases	0

dtype: int64

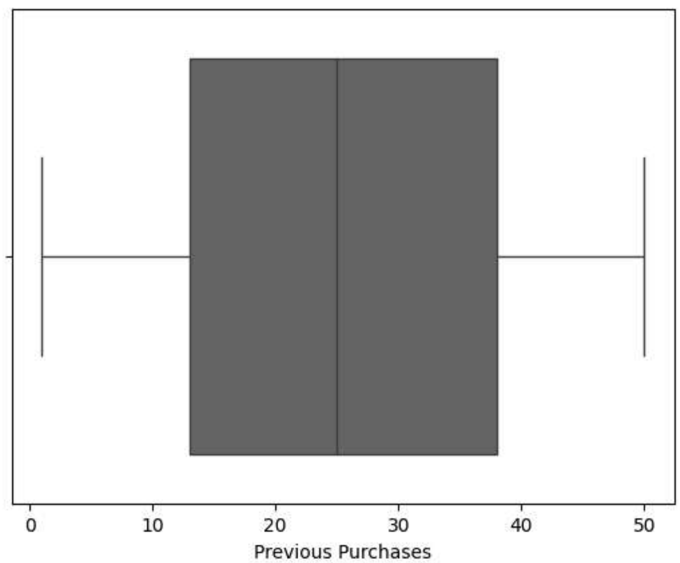
```
sns.boxplot(x=df['Purchase Amount (USD)'])
```

```
<Axes: xlabel='Purchase Amount (USD)'>
```




```python
sns.boxplot(x=df['Previous Purchases'])
```

```
<Axes: xlabel='Previous Purchases'>
```




```python
colx = ["Gender", "Category", "Location","Season", "Shipping Type", "Payment Method","Frequency of Purchases"]

for col in colx:
    unique_categories = pd.unique(df[col])

    print(f"{col}: {unique_categories}\n")
```

```
Gender: ['Male' 'Female']

Category: ['Clothing' 'Footwear' 'Outerwear' 'Accessories']

Location: ['Kentucky' 'Maine' 'Massachusetts' 'Rhode Island' 'Oregon' 'Wyoming'
 'Montana' 'Louisiana' 'West Virginia' 'Missouri' 'Arkansas' 'Hawaii'
 'Delaware' 'New Hampshire' 'New York' 'Alabama' 'Mississippi'
 'North Carolina' 'California' 'Oklahoma' 'Florida' 'Texas' 'Nevada'
 'Kansas' 'Colorado' 'North Dakota' 'Illinois' 'Indiana' 'Arizona'
 'Alaska' 'Tennessee' 'Ohio' 'New Jersey' 'Maryland' 'Vermont'
 'New Mexico' 'South Carolina' 'Idaho' 'Pennsylvania' 'Connecticut' 'Utah'
 'Virginia' 'Georgia' 'Nebraska' 'Iowa' 'South Dakota' 'Minnesota'
 'Washington' 'Wisconsin' 'Michigan']

Season: ['Winter' 'Spring' 'Summer' 'Fall']

Shipping Type: ['Express' 'Free Shipping' 'Next Day Air' 'Standard' '2-Day Shipping'
 'Store Pickup']
```

```
Payment Method: ['Venmo' 'Cash' 'Credit Card' 'PayPal' 'Bank Transfer' 'Debit Card']

Frequency of Purchases: ['Fortnightly' 'Weekly' 'Annually' 'Quarterly' 'Bi-Weekly' 'Monthly'
 'Every 3 Months']
```

```
df['Subscription Status'] = df['Subscription Status'].map({
    'Yes': 1,
    'No': 0
})

freq = {
   'Fortnightly':3,
   'Weekly':2,
   'Annually':7,
   'Quarterly':6,
   'Bi-Weekly':1,
   'Monthly':4,
   'Every 3 Months':5
}

df['Frequency of Purchases'] = df['Frequency of Purchases'].map(freq)

df['Discount Applied'] = df['Discount Applied'].map({
    'Yes': 1,
    'No': 0
})

df['Promo Code Used'] = df['Promo Code Used'].map({
    'Yes': 1,
    'No': 0
})
```

```
df.head(20)
```

	Customer ID	Age	Gender	Category	Purchase Amount (USD)	Location	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases
0	1	55	Male	Clothing	53	Kentucky	Winter	3.1	1	Express	1	1	14
1	2	19	Male	Clothing	64	Maine	Winter	3.1	1	Express	1	1	2
2	3	50	Male	Clothing	73	Massachusetts	Spring	3.1	1	Free Shipping	1	1	23
3	4	21	Male	Footwear	90	Rhode Island	Spring	3.5	1	Next Day Air	1	1	49
4	5	45	Male	Clothing	49	Oregon	Spring	2.7	1	Free Shipping	1	1	31
5	6	46	Male	Footwear	20	Wyoming	Summer	2.9	1	Standard	1	1	14
6	7	63	Male	Clothing	85	Montana	Fall	3.2	1	Free Shipping	1	1	49
7	8	27	Male	Clothing	34	Louisiana	Winter	3.2	1	Free Shipping	1	1	19
8	9	26	Male	Outerwear	97	West Virginia	Summer	2.6	1	Express	1	1	8
9	10	57	Male	Accessories	31	Missouri	Spring	4.8	1	2-Day Shipping	1	1	4
10	11	53	Male	Footwear	34	Arkansas	Fall	4.1	1	Store Pickup	1	1	26
11	12	30	Male	Clothing	68	Hawaii	Winter	4.9	1	Store Pickup	1	1	10
12	13	61	Male	Outerwear	72	Delaware	Winter	4.5	1	Express	1	1	37
13	14	65	Male	Clothing	51	New Hampshire	Spring	4.7	1	Express	1	1	31
14	15	64	Male	Outerwear	53	New York	Winter	4.7	1	Free Shipping	1	1	34
15	16	64	Male	Clothing	81	Rhode Island	Winter	2.8	1	Store Pickup	1	1	8
16	17	25	Male	Accessories	36	Alabama	Spring	4.1	1	Next Day Air	1	1	44
17	18	53	Male	Clothing	38	Mississippi	Winter	4.7	1	2-Day Shipping	1	1	36

Next steps: Generate code with df | New interactive sheet

CUSTOMER METRICS

1. Promo Dependency

```
df['Promo Dependency'] = (df['Discount Applied'] + df['Promo Code Used'])
df['Promo Dependency'] = df['Promo Dependency'].fillna(0)

df.sample(5)
```

	Customer ID	Age	Gender	Category	Purchase Amount (USD)	Location	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases
2993	2994	32	Female	Outerwear	21	Kansas	Spring	4.2	0	Free Shipping	0	0	18
923	924	55	Male	Clothing	24	Arkansas	Spring	4.2	1	2-Day Shipping	1	1	7
2010	2011	24	Male	Clothing	21	Texas	Summer	4.4	0	2-Day Shipping	0	0	3
374	375	69	Male	Accessories	78	Ohio	Spring	5.0	1	Store Pickup	1	1	39
321	322	41	Male	Footwear	36	Vermont	Summer	4.7	1	Free Shipping	1	1	48

2. Customer Lifetime Value (Proxy)

```
df['CLV Proxy'] = df['Purchase Amount (USD)'] * df['Frequency of Purchases'] * df['Previous Purchases']
df['CLV Proxy'] = df['CLV Proxy'].fillna(0)

df.sample(5)
```

	Customer ID	Age	Gender	Category	Purchase Amount (USD)	Location	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases
3138	3139	47	Female	Clothing	95	Kentucky	Fall	4.6	0	Next Day Air	0	0	40
2750	2751	51	Female	Footwear	66	Alaska	Spring	2.7	0	2-Day Shipping	0	0	26
3805	3806	48	Female	Accessories	57	New Jersey	Fall	3.1	0	Standard	0	0	31
1585	1586	37	Male	Clothing	25	Alabama	Fall	3.0	0	Next Day Air	1	1	40
1828	1829	33	Male	Outerwear	80	New Hampshire	Spring	3.9	0	Express	0	0	45

3. Loyal Customer Score A

```
df[['Previous Purchases',
    'Frequency of Purchases',
    'Subscription Status',
    'Purchase Amount (USD)',
    'Review Rating']].corr(numeric_only=True)
```

	Previous Purchases	Frequency of Purchases	Subscription Status	Purchase Amount (USD)	Review Rating
Previous Purchases	1.000000	0.005530	0.030859	0.008063	0.003568
Frequency of Purchases	0.005530	1.000000	-0.005068	0.004585	0.009100
Subscription Status	0.030859	-0.005068	1.000000	-0.006996	-0.002200
Purchase Amount	0.008063	0.004585	-0.006996	1.000000	0.029720

```
from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler(feature_range=(0,10))

df[['Previous Purchases Scaled','Frequency Scaled','Promo Dep Scaled','Subscription Scaled']] = scaler.fit_transform(df[['Previ
df.sample(5)
```

	Customer ID	Age	Gender	Category	Purchase Amount (USD)	Location	Season	Review Rating	Subscription Status	Shipping Type	...	Promo Code Used	Previous Purchases	Payme Metl
2271	2272	29	Male	Clothing	99	Alaska	Summer	3.4	0	Free Shipping	...	0	49	Pay
269	270	68	Male	Clothing	44	Georgia	Winter	3.3	1	Next Day Air	...	1	9	Ca
772	773	18	Male	Clothing	22	Illinois	Fall	3.6	1	Free Shipping	...	1	40	De C
1703	1704	39	Male	Accessories	22	California	Spring	4.0	0	Next Day Air	...	0	12	Ba Trans
945	946	38	Male	Footwear	38	Minnesota	Summer	4.1	1	Store Pickup	...	1	8	Ven

5 rows × 21 columns

```
df['Loyalty Score A'] =  ((df['Previous Purchases Scaled'] + df['Frequency Scaled'] + 10 - df['Promo Dep Scaled']) / 3).round(2
df.sample(5)
```

	Customer ID	Age	Gender	Category	Purchase Amount (USD)	Location	Season	Review Rating	Subscription Status	Shipping Type	...	Previous Purchases	Payment Method	F P
446	447	19	Male	Clothing	74	West Virginia	Summer	4.1	1	Store Pickup	...	15	Venmo	
533	534	57	Male	Footwear	60	North Carolina	Spring	4.5	1	Next Day Air	...	17	Debit Card	
2099	2100	31	Male	Clothing	78	Oklahoma	Summer	2.9	0	Express	...	50	PayPal	
1716	1717	69	Male	Clothing	47	Connecticut	Winter	4.7	0	Standard	...	30	Cash	
3041	3042	48	Female	Accessories	79	Rhode Island	Winter	4.0	0	Express	...	27	PayPal	

5 rows × 22 columns

```
df['Loyalty Score A'].describe()
```

	Loyalty Score A
count	3900.000000
mean	5.246746
std	2.205781
min	0.000000
25%	3.610000
50%	5.290000
75%	6.900000
max	10.000000

dtype: float64

4. Loyal Customer Score B

```
df['Loyalty Score B'] =  ((df['Previous Purchases Scaled'] + df['Frequency Scaled'] + df['Subscription Scaled']) / 3).round(2)
df.sample(5)
```

	Customer ID	Age	Gender	Category	Purchase Amount (USD)	Location	Season	Review Rating	Subscription Status	Shipping Type	...	Payment Method	Frequency of Purchases	Dep
745	746	46	Male	Clothing	92	Michigan	Fall	4.7	1	Standard	...	Venmo	4	
1310	1311	43	Male	Clothing	59	South Carolina	Winter	3.1	0	Store Pickup	...	Credit Card	5	
3135	3136	68	Female	Footwear	85	Indiana	Spring	4.7	0	Free Shipping	...	Venmo	3	
3635	3636	52	Female	Footwear	32	Colorado	Fall	3.3	0	Free Shipping	...	Credit Card	3	
2933	2934	27	Female	Accessories	52	Illinois	Summer	3.8	0	Standard	...	Venmo	4	

5 rows × 23 columns

Choosing best of score A or B

```
print(df[['Loyalty Score A','Loyalty Score B']].corr())
```

```
                 Loyalty Score A  Loyalty Score B
Loyalty Score A         1.000000         0.107913
Loyalty Score B         0.107913         1.000000
```

```
df[['Loyalty Score A','Loyalty Score B']].sample(10)
```

	Loyalty Score A	Loyalty Score B
1845	8.70	5.36
2108	6.31	2.98
79	5.08	8.41
259	4.15	7.48
2264	8.33	5.00
701	2.50	5.83
2901	9.17	5.84
2963	5.73	2.39
2728	4.98	1.65
1747	8.83	5.50

```
print(df['Loyalty Score A'].describe())
print(df['Loyalty Score B'].describe())
```

```
count    3900.000000
mean        5.246746
std         2.205781
min         0.000000
25%         3.610000
50%         5.290000
75%         6.900000
max        10.000000
Name: Loyalty Score A, dtype: float64
count    3900.000000
mean        4.246805
std         2.115668
min         0.000000
25%         2.700000
50%         4.070000
75%         5.652500
max        10.000000
Name: Loyalty Score B, dtype: float64
```

```
df.rename(columns={'Loyalty Score A': 'Loyalty Score'}, inplace=True)
df.drop(columns=['Loyalty Score B'], inplace=True)
df.head(5)
```

	Customer ID	Age	Gender	Category	Purchase Amount (USD)	Location	Season	Review Rating	Subscription Status	Shipping Type	...	Previous Purchases	Payment Method	Frequ Purcl
0	1	55	Male	Clothing	53	Kentucky	Winter	3.1	1	Express	...	14	Venmo	
1	2	19	Male	Clothing	64	Maine	Winter	3.1	1	Express	...	2	Cash	
2	3	50	Male	Clothing	73	Massachusetts	Spring	3.1	1	Free Shipping	...	23	Credit Card	
3	4	21	Male	Footwear	90	Rhode Island	Spring	3.5	1	Next Day Air	...	49	PayPal	
4	5	45	Male	Clothing	49	Oregon	Spring	2.7	1	Free Shipping	...	31	PayPal	

5 rows × 22 columns

5. Satisfaction Flag and CSAT (Proxy)

```
df['Review Rating'].describe()
```

	Review Rating
count	3900.000000
mean	3.750064
std	0.713573
min	2.500000
25%	3.100000
50%	3.750000
75%	4.400000
max	5.000000

dtype: float64

```
df['Satisfaction Tier'] = (df['Review Rating'] > df['Review Rating'].mean()).astype(int)

csat_proxy = df['Satisfaction Tier'].mean()*100
# csat_proxy = ((df['Satisfaction Tier']==1).sum()/ df['Customer ID'].count())*100
print(csat_proxy)

df[['Review Rating', 'Satisfaction Tier']].sample(5)
```

```
49.666666666666664
```

	Review Rating	Satisfaction Tier
1961	2.6	0
2188	4.7	1
3025	3.4	0
2092	2.6	0
3200	4.8	1

Customer Segmentation

```
clv_cutoff = df['CLV Proxy'].median()
loyalty_cutoff = 7

conditions = [
    (df['CLV Proxy'] >= clv_cutoff) & (df['Loyalty Score'] >= loyalty_cutoff),
    (df['CLV Proxy'] >= clv_cutoff) & (df['Loyalty Score'] < loyalty_cutoff),
    (df['CLV Proxy'] < clv_cutoff) & (df['Loyalty Score'] >= loyalty_cutoff),
    (df['CLV Proxy'] < clv_cutoff) & (df['Loyalty Score'] < loyalty_cutoff)
]
```

```
    choices = [
        'Platinum Champions',
        'At-Risk High-Value',
        'Emerging Devotees',
        'Low-Engagement Casuals'
    ]

    df['Customer Segment'] = np.select(conditions, choices, default = 'Others')
```

```
df['Customer Segment'].value_counts()
```

Customer Segment	count
Low-Engagement Casuals	1855
At-Risk High-Value	1118
Platinum Champions	834
Emerging Devotees	93

dtype: int64

Segment Profiling

Demographic profile ⟶ Age, Gender

Behavioral profile ⟶ Purchase Amount, Previous Purchases, Promo Dependency

Product preference ⟶ Category

Payment preference ⟶ Payment Method

Geographic profile ⟶ Location